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Performance Evaluation of the Andasol-1 Solar Power Plant in Algerian Climatic Conditions

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Abstract: The generation of electricity from renewable sources is imperative in order to fight against the environmental and economic problems associated with fossil fuel utilization. The entire world energy sector is experiencing a rapid transformation towards renewable and sustainable energy sources, and a promising technological advancement in this regard is concentrated solar power (CSP), with a notable example being parabolic trough solar power plants. In this paper, we investigate the operational efficiency of the Andasol-1 parabolic trough solar power plant with 50 MWe output power in various Algerian desert locations, i.e., the Béchar, Ghardaïa, and Tamanrasset regions. A thorough examination of the Andasol-1 plant’s technical characteristics is made, and an extensive economic and energy analysis is conducted based on System Advisor Model (SAM) software. Our study results reveal that the Tamanrasset region seems to be the most suitable area